

Master of Science in Computer Engineering Specialization: Data Science and Analytics

Course Title : **Foundation of Data Science & Analytics**

Course Code :

Course Type : **Core**

Lecture : 4 (hr. per week)

Tutorial : 0 (hr. per week)

Practical : 0 (hr. per week)

Year : I

Part : I

Course Objective:

The objective of the course is to provide the fundamental principles of Data Science and analytics specifics to the students. The course also provides the basic concepts on association analysis, model building processes steps and matrix factorization methods. In addition to the required basic mathematics and exploratory data analysis related statistics, the course also deals with DBMS, relational algebra, SQL and NoSQLs.

Course Outline:

- 1. Introduction to Data Science (2 Hrs)**
Data Science Hype, Why data science, Getting Past the Hype, The Current Landscape, Role of Data Scientist
- 2. Data Types and Data Science Processes (7 Hrs)**
Facets of data: Structured data, Unstructured data, Natural language, Machine-generated data , Graph-based or network data, Audio, image, and video, Streaming data
Process Overview, Defining goals, Retrieving data, Data preparation, Exploratory Data Analysis, Data Wrangling & Cleaning, Data Integration and Transformation, Data Reduction, Data modeling and Result Presentation
- 3. Mathematical Foundation for Data Science (10 hours)**
Introduction and Descriptive Statistics : An overview of probability and statistics, Pictorial and tabular methods in descriptive statistics, Measures of central tendency, dispersion, and direction, Joint and conditional probabilities (3 Hrs)
Random Variables and Probability Distributions: Random variables, Probability distributions for random variables, Expected values of discrete random variables and continuous distributions, The binomial probability distribution, The Poisson probability distribution (3 Hrs)
Central limit theorem and Sample distribution concepts, Normal approximation; Hypothesis Testing Procedures: Tests about the mean of a normal population, The t-test, Z-tests for differences between two populations means, The two-sample t-test, Confidence interval for mean of normal population (4 Hrs)
- 4. Regression and associated Models (11 Hrs)**
Empirical Models, Simple Linear Regression, MLE and Least Square Estimator, Logistic Regression, Hypothesis tests in simple linear regression, t-tests and ANOVA, Confidence intervals, Residual Analysis, Coefficient of Determination, Correlation (3 Hrs)
Multiple Linear Regression, Matrix approach to Multiple Linear Regression, Polynomial Regression Models, Categorical Regressors, Indicator variables, Selection of variables and Model building (4 Hrs)
Matrix Factorization (MF), Probabilistic and, Non-Negative MF, Industry Applications (4 Hrs)

5. **Modeling and validation processes for Machine Learning Techniques** (8 Hrs)
Supervised learning algorithms & Unsupervised learning algorithms.
Modeling Process, Training /Validating model, Cross Validation methods, Predicting new observations Interpretation
Measures for Model Performance and Evaluation: Classification accuracy, Confusion matrix, Sensitivity, Specificity, Precision, Recall, F-score, ROC curve, Clustering performance measures, other measures
6. **Association and Other types of Analysis** (12 Hrs)
Market Basket Analysis using frequent itemset, Association rules generation from transactional dataset, Apriori and other algorithms, Correlation analysis
Outlier Analysis, Trend analysis, Time series analysis, Social network analysis
7. **Database and Datawarehousing** (6 Hrs)
DBMS fundamentals, Relational Algebra and SQL, OLTP, Datawarehouse, Multidimensional data model, Data Cubes, NoSQL, OLAP Operations
8. **Ethics and Recent Trends** (4 hours)
Data Science Ethics, Doing good data science, Owners of the data, Privacy aspects, Social impact, Getting informed consent, The Five Cs, Future Trends

Evaluation Scheme:

a. Internal Examination

Type	Weightage
Minor tests	50%
Assignments	50%

b. Final Examination

There will be five units of questions carrying 12 marks each. The question will cover all chapters of the syllabus. The evaluation scheme will be as indicated in the table:

Units	Chapters	Marks
1	1, 3	12
2	2, 7	12
3	4	12
4	5, 8	12
5	6	12
Total		60

References:

1. Introducing Data Science: Big Data. Machine Learning and More, Using Python Tools. Cielen D, Meysman AD, Ali M. Manning, 2016
2. An Introduction to Statistical Learning: with Applications in R, Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Springer, 1st edition, 2013
3. Applied Statistics and Probability for Engineers, Douglas C. Montgomery, George C Runger, Wiley, 2014
4. Ethics and Data Science, D J Patil, Hilary Mason, Mike Loukides, O' Reilly, 2018
5. Applied Data Science with Python and Jupyter: Galea A., Packt Publishing Ltd; 2018.
6. Adhikari A, DeNero J. Computational and Inferential Thinking: The Foundations of Data Science., 2017

Institute of Engineering
Pulchowk Campus
Department of Electronics and Computer Engineering

Master of Science in Computer Engineering Specialization: Data Science and Analytics

Course Title : **Machine Learning and Computational Intelligence**

Course Code :

Course Type : **Core**

Lecture : 4 (hr. per week)

Tutorial : 0 (hr. per week)

Practical : 0 (hr. per week)

Year : I

Part : I

Course Objective:

The objective of this course is to learn the fundamentals of pattern recognition and its relevance to classical and modern problems. It also talks about to identify where, when and how pattern recognition can be applied. The students will learn how to use pattern recognition in the real world. Students are introduced to recent applications of pattern recognition in the fields of science, Engineering, medicine, cognitive science and bioinformatics.

Course Outline:

1. Introduction

(5 hours)

Artificial Intelligence, Machine learning, Computational Intelligence, History, Computational Intelligence Paradigms; Data and tools, review of statistics, training, validation and test data, theory of learning – feasibility of learning – error and noise – training versus testing, generalization bound – approximation-generalization tradeoff – bias and variance – learning curve

2. Feature Extration

(8 hours)

Feature Selection: Data Preprocessing, Class Separability Measures, Feature Subset Selection, Bayesian Information Criterion
Dimensionality Reduction: Basis Vectors, Singular Value Decomposition, Independent Component Analysis, Kernel PCA, Wavelets
Additional Features and Template Matching: Texture, Shape and Size Characterization, Fractals, Features for Audio, Template Matching Using Dynamic Time Warping and Edit Distance

3. Bayes Decision Theory

(5 Hrs)

Discriminant Functions and Services, the Normal Distribution, Bayesian Classification, Estimating Probability Density Functions, Nearest Neighbor Rules, Bayesian Networks

4. Neural Networks

(5 Hrs)

Basic Neuron, Perceptrons, Hebbian Learning, Adaline Network , Application of Neural Networks

5. Classification

(10Hrs)

Linear Classifiers: the Perceptron Algorithm, Least-Squares Methods; Nonlinear Classifiers: Multilayer Perceptron's, Back Propagation Algorithm, Decision Trees, Combinations of Classifiers, Boosting
Context Dependent Classification; SVM and Other methods

- 6. Clustering (10Hrs)**
Sequential Algorithms, Hierarchical Algorithms, Functional Optimization-Based Clustering, Graph Clustering, Learning Clustering, Clustering High Dimensional Data, Subspace Clustering, Cluster Validity Measures
- 7. Deep Learning (9 Hrs)**
Intorduction, Architectures, Hyperparameter tuning, Fundamentals on various networks like CNN, RNN, GRU,SNN, GAN e.t.c
- 8. Evolutionary Computation & Intelligence (6Hrs)**
Genetic Algorithm, Particle Swarm Optimization, Ant Colony Optimization, Fuzzy Logic
- 9 Performance Evaluation and Measures (4 Hrs)**
Classification accuracy, Confusion matrix Misclassification costs, Sensitivity and specificity, ROC curve, Recall and precision, box plot, confidence interval. Estimating Performance, Reliability of quality estimations Confidence interval, Cross validation Data visualization

Evaluation Scheme:

a. Internal Examination

Type	Weightage
Minor tests	50%
Assignments	50%

b. Final Examination

There will be five units of questions carrying 12 marks each. The question will cover all chapters of the syllabus. The evaluation scheme will be as indicated in the table:

Units	Chapters	Marks
1	1, 8	12
2	2, 9	12
3	3, 5	12
4	4, 7	12
5	6	12
Total		60

References:

1. Engelbrecht AP. Computational intelligence: an introduction. John Wiley & Sons; 2007.
2. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer.
3. Introduction to Machine Learning with Python: A Guide for Data Scientists, 1st ed., O'Reilly Media
4. David Barber, Bayesian Reasoning and Machine learning, Oxford University press
5. Tom M Mitchell, Machine Learning, First edition, McGraw Hill Education
6. Simon Haykin, “Neural Networks: A Comprehensive Foundation”, PearsonEducation Asia, 2001.

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Pulchowk Campus
Department of Electronics and Computer Engineering

Master of Science in Computer Engineering Specialization: Data Science and Analytics

Course Title : **Big Data Analytics**

Course Code :

Course Type : **Core**

Lecture : 4 (hr. per week)

Tutorial : 0 (hr. per week)

Practical : 0 (hr. per week)

Year : I

Part : I

Course Objective:

The objective of the course is to familiarize the students with big data and latest trends in big data analytics with the introduction of various technologies for handling big data including Hadoop and components in Hadoop Platform. After completing the course, students will be able to perform basic exploration of large, complex datasets and scalable big data analysis. In addition, the students will be also able to apply big data tool for advanced analytics disciplines such as predictive analytics, data mining, text analytics and statistical analysis.

Course Outline:

- 1. Fundamentals of Big Data Analytics (4 hours)**
Big Data and the V's of Big Data, Handling and Processing Big Data, The Big Data landscape, Big Data Analytics, Examples of real world big data problems
- 2. Technologies for Handling Big Data (8 hours)**
GFS, HDFS, Google Big Table, Introduction to Hadoop, functioning of Hadoop, MapReduce, RDDs Cloud Computing for big data
- 3. Understanding Big Data Technology Foundations (10 hours)**
Big data stack i.e. data source layer, ingestion layer, storage layer, processing layer, security layer, visualization layer, visualization approaches; Architectural design patterns and programming models used for Real-World Applications, Lambda Architecture
- 4. Understanding Big data Ecosystem (12hours)**
Hadoop and its ecosystem, Introduction and Experimentation with Apache Flume, Apache Kafka, Apache Zookeeper, Apache Spark, Apache Mesos, Apache Kudu; Amazon and Google Cloud Platform, Microsoft Azure , Amazon Kinesis
- 5. Using Big Data for Analytics (10 hours)**
Basic approaches to querying and exploring big data; New Databases for Big data Analytics - Classification, Characteristics and Comparison ; Apache HBase, Apache Hive, Apache Cassandra; Descriptive, Diagnostic, Predictive, Prescriptive Analytics, Stream Analytics and Location Analytics; Case studies for big data analytics

6. Machine Learning with Big Data**(12 hours)**

Introduction to parallel, distributed and scalable machine learning; Using processing layer tools (Apache Spark, Apache Mahout) to train, evaluate, and validate basic predictive models.
Case studies for application of machine learning in big data.

7. Latest Trends and Research on Big Data Issues**(4 hours)****Evaluation Scheme:****a. Internal Examination**

Type	Weightage
Minor tests	50%
Assignments	50%

b. Final Examination

There will be five units of questions carrying 12 marks each. The question will cover all chapters of the syllabus. The evaluation scheme will be as indicated in the table:

Units	Chapters	Marks
1	1, 2	12
2	3	12
3	4	12
4	5, 7	12
5	6	12
Total		60

References:

1. Tom White. Hadoop: The Definitive Guide, Storage and Analysis at Internet Scale, O'Reilly Media, 4th Edition, 2015
2. Nathan Marz and James Warren. Big Data: Principles and best practices of scalable realtime data systems, Manning Publications, First Edition, 2015
3. Mark Grover, Ted Malaska, Jonathan Seidman, Gwen Shapira. Hadoop Application Architectures: Designing Real-World Big Data Applications, O'Reilly Media, First Edition, 2015
4. Holden Karau, Andy Konwinski, Patrick Wendell, Matei Zaharia. Learning Spark: Lightning -Fast Big Data Analysis, O'Reilly Media, 2015
5. Nataraj Dasgupta, Practical Big Data Analytics: Hands-on techniques to implement enterprise analytics and machine learning using Hadoop, Spark, NoSQL and R, Packt Publishing, 2018

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Master of Science in Computer Engineering Specialization: Data Science and Analytics

Course Title : **Distributed and Edge Computing**

Course Code :

Course Type : **Core**

Lecture : 4 (hr. per week)

Tutorial : 0 (hr. per week)

Practical : 0 (hr. per week)

Year : I

Part : I

Course Objective:

The objective of the course is to familiarize students with current trends and principles in distributed and edge computing systems. After completing the course students will be able to analyze distributed and edge computing systems, design and implement modern and future systems and use their knowledge in analysis and research work in distributed and edge computing systems.

Course Outline:

1. Introduction

4 hrs

Basics of Distributed Systems, Evolution of Distributed Computing Systems, Distributed System Models and Enabling Technologies, Distributed system architecture, Issues in Design of Distributed Systems, Types of Distributed System, Distributed Operating Systems, Network Operating Systems, Distributed Web Based System.

2. Message Passing and Communication

10 hrs

Communications in Distributed System, Layered Network Protocols, Middleware, Invocation methods in Distributed system , Message Oriented Communication, RPC Messages, Marshaling Arguments and Results, Communication Protocols for RPC, Code Migration, Software Agents.

3. Synchronization

10 hrs

Clock Synchronization, Physical/Logical Clocks, Mutual Exclusion, Global State, State Recording, Election Algorithms.

4. Replication

10 hrs

Reason for Replication, Continuous Consistency, Consistent Ordering of Operations, Client Centric Consistency, Replica Management, Consistency Protocols, concurrency control, exception handling, load balancing and sharing approach, Fault Tolerance System, Failure Models, Process Resilience, High Available System, recovery approaches.

5. Distributed Computation using Cloud Computing

10 hrs

Basics of Cloud Computing, Benefits of Cloud Computing, Cloud Characteristics, Service Oriented Architecture, REST, Layered Cloud Architecture Design, NIST Cloud Computing Reference Architecture, IaaS, PaaS, SaaS, Consensus and the Paxos algorithm

6. Edge Computing and Emerging Trends**8 hrs**

Basics of Edge Computing, Cloud Computing vs Edge Computing, Fog Computing, IoT, Opportunities and Challenges of Edge Computing, Integrating Edge, Fog & Cloud Infrastructures, Network Slicing, Middleware for Fog and Edge Computing.

7. Distributed Systems Security**8 hrs**

Overview of Cryptography and data privacy, Security issues and techniques in distributed systems, Security challenges in edge computing, Introduction to digital forensics.

Evaluation Scheme:**a. Internal Examination**

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Minor tests	50%
Assignments	50%

b. Final Examination

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Units	Chapters	Marks
1	1, 5	12
2	2	12
3	3	12
4	4	12
5	6,7	12
Total		60

References:

1. Andrews S. Tanenbaum, Marteen V. Steen, “Distributed Systems Principles and Paradigms”, 2nd Edition, PHI
2. Raj Kumar Buyya, Satish Narayana Srirama “Fog and Edge Computing”, JohnWiley & Sons
3. Raj Kumar Buyya, Christian Vecchiola, S. Thamarai Selvi, “Mastering Cloud Computing, Foundations and Applications Programming”
4. George Coulouris, Jean Dollimore, Tim Kindberg, Gordon Blair, “Distributed Systems: Concepts and Design” 5th Edition, Pearson
5. Kenneth P Birman, “Guide to Reliable Distributed Systems: Building High-Assurance Applications and Cloud-Hosted Services” Springer
6. Sukumar Ghosh, “Distributed Systems: An Algorithmic Approach” 2nd Ed., Chapman & Hall CRC.